requirements analysis, helping you focus on the capabilities and features of the application that will genuinely add value to the business.

A key part of this practice involves defining scenarios, or concrete examples of how a particular feature or story works. These scenarios will help you to validate and extend your understanding of the problem, and they're also an excellent communication tool. They act as the foundation of the acceptance criteria, which you then integrate into the build process in the form of automated acceptance tests. In conjunction with the automated acceptance tests, these examples guide the development process, helping designers to prepare effective and functional user-interface designs and assisting developers to discover the underlying behaviors that they'll need to implement to deliver the required features.

In the rest of this chapter, we'll look at a practical example of this process in action. We'll touch on aspects of the whole development cycle, from business analysis to implementing, testing, and maintaining the code.

2.1 Introducing the train timetable application

For this chapter's example, suppose you work for a large, government public-transport department. You've been asked to lead a small team building a service that will provide train timetable data and real-time updates about delays, track work, and so on for various mobile apps used by commuters. Figure 2.1 illustrates the rail network you'll be working with.

The department has just introduced Agile and BDD practices, so you'll start by talking to the key stakeholders to make sure that you and your team have a clear idea of the business goals driving the project. This will help the team deliver a better, more targeted application.

When you've understood and articulated the business goals, you'll need to work with your business analyst and business stakeholders to decide what software features will be able to achieve these goals. These features are high-level requirements, such as "Provide travellers with optimal itineraries between stations" or "Notify commuters if their train is late."

You probably won't be able to deliver features this big in one go, so you'll need to break them down into smaller units, known to Agile practitioners as *stories*. These stories might include things like "Find the optimal itinerary between stations on the same line" and "Find the optimal itinerary between stations on different lines."

When it comes to implementing a story, you get together with your business analyst, developer, and tester to describe the story in terms of concrete examples. Many of these examples will already have been discussed with the business stakeholders. These examples become the *acceptance criteria* for the story, and they're expressed in a formal BDD style that you can later automate:

```
Given Western line trains leave Parramatta at 7:58, 8:02, 8:08, 8:11
When I want to travel from Parramatta to Town Hall at 8:00
Then I should be told to take the 8:02 train
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